Rotational Diagonal Kalman Linear Attention: Model, Lessons from Mamba-2/Mamba-3, and Production Implementation

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Abstract

Mamba-2 supplies a strong selective state-space baseline, but its SISO state transition is a scalar radial contraction: it cannot rotate a two-dimensional latent subspace. The installed Mamba-3 SISO implementation adds rotation economically by accumulating a bounded angular rate and applying the same rotary transform to queries and keys. Its most useful lesson is therefore not a particular state-space write rule, but a representation: pairwise rotation can be moved out of the recurrent state update and into a phase-rotated query/key basis.

Rotational diagonal Kalman linear attention (RDiag KLA, E2 clock) adopts that representation while retaining KLA's diagonal Kalman covariance and innovation write. It pair-ties radial transition and process-noise parameters, uses a detached live radial statistic to set a bounded token/head clock, scans phase in a documented negative gauge, and isolates the phase VJP from radial dynamics. This note defines the model, the precise finite-precision implementation contract, and the tests used for promotion. The sealed E2 baseline implementation is commit `c3346feaecae1486cdbc71f82b4edeacc4198950`; its immutable production snapshot passed the complete one-H200 correctness and performance gate (117 core RDiag tests plus 54 E2 tests). Distributed training-system checks and the multi-node soak/attestation remain before the planned 750M DiagKLA production submission. The one-H200 result concerns implementation correctness and measured kernel/layer performance, not the eventual language-model quality of a long training run. A newer shared-DT investigation, including its unresolved production-qualification status, is reported in Section 10.

1 Scope and notation

Let B, T, H, K, V, D denote batch size, sequence length, number of heads, key dimension per head, value dimension per head, and model width. The rotary prefix has even width $K_r = 2P$, where P is its number of adjacent pairs. Tensors use the following shapes:

$$\begin{aligned} &\boldsymbol{x} \in \mathbb{R}^{B\times T\times D}, && \boldsymbol{q}, \boldsymbol{k}, \boldsymbol{\alpha}, \boldsymbol{\omega} \in \mathbb{R}^{B\times T\times H\times K},\\ &\boldsymbol{v} \in \mathbb{R}^{B\times T\times H\times V}, && \boldsymbol{e} \in \mathbb{R}^{B\times T\times P},\\ &\boldsymbol{\tau} \in \mathbb{R}^{B\times T\times H}, && \boldsymbol{\delta}, \boldsymbol{\phi} \in \mathbb{R}^{B\times T\times H\times P}. \end{aligned}$$

The raw angle e_{btj} is shared across heads; the clock τ_{bth} makes the resulting increment head-specific. For an adjacent pair $(2j, 2j+1)$, define

$$R(\vartheta) = \begin{bmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{bmatrix}, \qquad R(\vartheta)^{\mathsf{T}} R(\psi) = R(\psi - \vartheta). \tag{1}$$

All equations below are per batch element unless a batch index is useful. "SISO" follows the Mamba implementation's terminology; a head can still carry multiple value channels that share the same scalar state-space coefficients.

2 What Mamba-2 has—and what it lacks

For head h, value channel p, and SSM state coordinate n, the Mamba-2 selective SISO update can be written

$$d_{th} = \operatorname{softplus}(\widetilde{d}_{th} + b_h^d), \qquad a_{th} = \exp(A_h d_{th}), \qquad A_h < 0, \tag{2}$$

$$X_{t,h,p,n} = a_{th} X_{t-1,h,p,n} + d_{th} B_{t,n} u_{t,h,p}, \qquad y_{t,h,p} = \sum_n C_{t,n} X_{t,h,p,n} + D_h u_{t,h,p}. \tag{3}$$

The input projection is ordered as $[z, x, B, C, dt]$. It emits one dt row per token and head; B and C are token-dependent state vectors. The factor a_{th} is a scalar radial contraction over the head state, and $d_{th} B_t u_t$ is its discretized injection. There is no angle projection, phase state, adjacent-pair structure, sine/cosine operation, or 2×2 rotation in this path. Mamba-2 therefore supplies selective decay and injection, but not oscillatory transport. A pair of real coordinates within a head shares the same scalar decay, but that factor is $a_{th} I$: it has no skew component, phase state, or mechanism for rotating one coordinate into the other.

3 The installed Mamba-3 SISO rotational path

This section describes the code shipped in the installed `mamba-ssm 2.3.2.post1` package, not an inferred variant. Its input projection is ordered

$$[z, x, B, C, dd_{dt}, dd_A, \text{trap}, \text{angle}]. \tag{4}$$

The dd_{dt}, dd_A, and trapezoid fields each have H rows. In contrast, the angle field has only P_M rows, where P_M is half of the selected half- or full-rotary state prefix. The raw angle rows are expanded across heads. Thus Mamba-3 separates a token/head clock from token/pair angular content:

$$DT_{th} = \operatorname{softplus}(dd_{dt,th} + b_h^d), \tag{5}$$

$$A_{th} = \min\{-\operatorname{softplus}(dd_{A,th}), -A_{\text{floor}}\}, \qquad \rho_{th} = \exp(A_{th} DT_{th}), \tag{6}$$

$$\nu_{tj} = \pi \tanh(e_{tj}), \tag{7}$$

$$\Delta\theta_{thj} = DT_{th} \nu_{tj}, \tag{8}$$

$$\theta_{thj} = \theta_{0,hj} + \sum_{s=1}^{t} \Delta\theta_{shj} pmod 2\pi. \tag{9}$$

The cumulative phase is positive. For every selected adjacent pair, the SISO kernel applies the same $R(+\theta_{thj})$ to Q_t and K_t. Consequently the same-token dot product is unchanged, whereas a cross-token term obtains the relative rotation

$$(R(\theta_t) q_t)^\top R(\theta_s) k_s = q_t^\top R(\theta_s - \theta_t) k_s. \tag{10}$$

If S_t denotes a state in a fixed physical frame, this is equivalent to the moving-frame recurrence

$$S_t^- = \rho_t R(-\Delta\theta_t) S_{t-1}, \tag{11}$$

pair by pair, with scalar radial decay ρ_t. The identity follows from $R(\theta_t)^\top R(\theta_{t-1}) = R(-\Delta\theta_t)$. It is the key computational idea: no dense rotational state transition is needed when radial coefficients are pair-isotropic.

Mamba-3 additionally uses a trapezoidal SSM write. With

$$\gamma_t = DT_t \,\mathrm{sigmoid}(\mathrm{trap}_t), \qquad \eta_{t+1} = DT_{t+1}\big(1 - \mathrm{sigmoid}(\mathrm{trap}_{t+1})\big), \tag{12}$$

the offline kernel scales token t's rotated key by $\gamma_t + \eta_{t+1}$, while its same-token contribution uses γ_t. At a streaming boundary, the cached previous key/value contributes the missing η_t term. This is a two-endpoint discretization of the Mamba write; it is not the KLA innovation update defined below.

The angle clock is also coupled in backward. The VJP of the positive cumulative phase is a reverse cumulative sum. It produces both

$$\bar{e}_{tj} = \sum_h \overline{\Delta\theta}_{thj}\, DT_{th}\, \pi\big(1 - \tanh^2(e_{tj})\big), \tag{13}$$

$$\overline{DT}^{\mathrm{angle}}_{th} = \sum_j \overline{\Delta\theta}_{thj}\, \pi \tanh(e_{tj}). \tag{14}$$

Here the sum over h is the backward of the head expansion of the shared raw angle rows. The combined implementation explicitly adds $\overline{DT}^{\mathrm{angle}}$ to the DT gradient from the radial SSM dynamics. This is a coherent choice for a shared physical discretization clock, but it is intentionally not the RDiag E2 gradient contract.

4 RDiag KLA with the E2 bounded clock

4.1 Pair-tied radial dynamics

Let the radial preactivation be f_{bthk} with bias b^d_{hk}, and let

$$d_{bthk} = \mathrm{softplus}(f_{bthk} + b^d_{hk}). \tag{15}$$

For rotary pair j, define the pair step and tied process noise

$$\bar{d}_{bthj} = \tfrac{1}{2}(d_{bth,2j} + d_{bth,2j+1}), \tag{16}$$

$$\bar{\omega}_{bthj} = \tfrac{1}{2}(\omega^{\mathrm{in}}_{bth,2j} + \omega^{\mathrm{in}}_{bth,2j+1}), \tag{17}$$

$$\alpha_{bth,2j} = \alpha_{bth,2j+1} = \exp\Big[-\exp(A^{\log}_h)\bar{d}_{bthj}\Big], \tag{18}$$

$$\omega_{bth,2j} = \omega_{bth,2j+1} = \bar{\omega}_{bthj}. \tag{19}$$

Only the selected prefix is pair-tied; tail lanes retain the diagonal-KLA frontend's original α and ω. Pair tying is not cosmetic. On each pair it makes $\mathrm{diag}(\alpha_t) = \alpha_{tj} I_2$ and the diagonal process covariance $\mathrm{diag}(\omega_t) = \omega_{tj} I_2$, so the *prediction* terms commute with $R(\vartheta)$. It does not force posterior information or the Kalman gain to be pair-isotropic; an observation generally splits them. The diagonal posterior approximation is defined in the current moving frame, as made explicit below.

4.2 Detached live radial clock and persistent anchor

Set the open upper clock bound to $C = 0.1$. At construction, snapshot one FP32 anchor per head,

$$a_h = \mathrm{mean}_k\, \mathrm{softplus}(b^d_{hk}), \qquad 0 < a_h < C. \tag{20}$$

The anchor is a persistent buffer, not a parameter. At each token compute the detached live radial mean

$$m_{bth} = \mathrm{mean}_k \,\mathrm{softplus}\Big(\mathrm{stopgrad}(f_{bthk} + b^d_{hk})\Big), \tag{21}$$

and a learnable token/head residual

$$z^{\mathrm{clk}}_{bth} = \frac{\langle w_h, x_{bt}\rangle}{\sqrt{D}} + \beta_h, \tag{22}$$

evaluated in FP32. The E2 clock is

$$\boxed{\tau_{bth} = C\ \mathrm{sigmoid}\Big[\log m_{bth} - \log(C - a_h) + \tanh(z^{\mathrm{clk}}_{bth})\Big]} \qquad (C = 0.1). \tag{23}$$

The bounded residual is strictly in $(-1, 1)$ and the sigmoid keeps a finite valid clock strictly in $(0, C)$. The implementation does not clamp or use a straight-through estimator: invalid, nonfinite, endpoint, or dead-derivative conditions are made visible to health checks, with invalid clock outputs replaced by NaNs.

Detachment is a model choice, not merely an optimization. The current radial timescale determines the forward phase scale, while the phase loss cannot change f, b^d, $A^{\log}$, α, or ω through this clock. The persistent anchor fixes the initialization reference across save/resume; recomputing it from a trained b^d would change the model after restart.

4.3 Angle, sign convention, and rotating basis

One projection shared across heads produces e_{btj}. The head-specific increment is

$$\delta_{bthj} = \pi \tanh(e_{btj})\tau_{bth}, \qquad |\delta_{bthj}| < 0.1\pi. \tag{24}$$

RDiag uses the negative inclusive phase scan

$$\phi_{bthj} = \phi_{b0,hj} - \sum_{s=1}^{t} \delta_{bshj}, \tag{25}$$

implemented in FP32 and reduced to $[0, 2\pi)$ at the bounded-clock consumer. The same $R(+\phi_{thj})$ is applied to the corresponding pairs of q_t and k_t. Hence, for $s < t$,

$$(R(\phi_t)q_t)^{\mathsf{T}} R(\phi_s)k_s = q_t^{\mathsf{T}} R\left(\sum_{u=s+1}^{t} \delta_u\right) k_s. \tag{26}$$

Under the column-vector convention of Eq. (1), the negative scan therefore represents $R(+\delta_t)$ transport. Mamba-3's positive scan represents $R(-\Delta\theta_t)$. This sign is a gauge choice because replacing the zero-initialized angle projection by its negative maps one convention to the other. It is nevertheless an implementation invariant: forward, streaming carry, and the negative suffix-scan VJP must use the same sign.

More explicitly, let $\mathcal{R}_t$ be block diagonal with the pair rotations $R(\phi_{thj})$ and an identity tail. The corresponding physical transition and process covariance are

$$D_t = \mathcal{R}_t^{\mathsf{T}} \,\mathrm{diag}(\alpha_t)\mathcal{R}_{t-1}, \qquad \Omega^{\mathrm{phys}}_t = \mathcal{R}_t^{\mathsf{T}} \,\mathrm{diag}(\omega_t)\mathcal{R}_t. \tag{27}$$

On a tied pair, these reduce to $D_{tj} = \alpha_{tj} R(+\delta_{tj})$ and $\Omega^{\mathrm{phys}}_{tj} = \omega_{tj} I_2$. With moving-frame state $\bar{S}_t = \mathcal{R}_t S_t$ and addresses $\bar{k}_t = \mathcal{R}_t k_t$, $\bar{q}_t = \mathcal{R}_t q_t$, the mean-state recurrence becomes purely radial and can reuse the diagonal-KLA memory kernel.

4.4 Diagonal Kalman gain and innovation memory

Let the phase-rotated addresses be $\widetilde{q}_t = \bar{q}_t$ and $\widetilde{k}_t = \bar{k}_t$. With posterior diagonal information $c_{t-1,hk} > 0$, observation noise $r_{th} > 0$, and information scale $s_h > 0$, RDiag KLA uses the moving-frame state (written as S_t below for brevity)

$$p_{thk}^{-} = \frac{\alpha_{thk}^2}{c_{t-1,hk}} + \omega_{thk}, \tag{28}$$

$$\kappa_{thk} = \frac{p_{thk}^{-}\widetilde{k}_{thk}}{r_{th} + \sum_i p_{thi}^{-}\widetilde{k}_{thi}^2}, \tag{29}$$

$$c_{thk} = \frac{1}{p_{thk}^{-}} + s_h \frac{\widetilde{k}_{thk}^2}{r_{th}}. \tag{30}$$

For associative state $S_{th} \in \mathbb{R}^{K \times V}$,

$$S_{th}^{-} = \operatorname{diag}(\alpha_{th}) S_{t-1,h}, \tag{31}$$

$$S_{th} = S_{th}^{-} + \boldsymbol{\kappa}_{th} \left(\boldsymbol{v}_{th} - \widetilde{\boldsymbol{k}}_{th}^{\mathsf{T}} S_{th}^{-} \right)^{\mathsf{T}}, \tag{32}$$

$$o_{th} = S_{th}^{\mathsf{T}} \left(K^{-1/2} \widetilde{\boldsymbol{q}}_{th} \right). \tag{33}$$

This is the KLA estimator. It is not a Mamba SSM write: κ_t is an adaptive anisotropic Kalman gain and the write is the current innovation. The diagonal information recurrence can be computed by its existing associative Möbius scan, followed by the existing chunked affine memory recurrence. In physical coordinates, c_t represents the model's diagonal projection of $\mathcal{R}_t P_t^{-1} \mathcal{R}_t^{\mathsf{T}}$; it is not a claim that rotating an arbitrary anisotropic posterior covariance remains diagonal. In particular, c, κ, k, and q are never pair-tied.

4.5 VJP isolation

The computational graph has two intentionally separate branches:

$$\begin{aligned} (f, b^d, A^{\log}, \omega^{\text{in}}) &\longrightarrow (\alpha, \omega) \longrightarrow \text{Kalman dynamics}, \\ \operatorname{stopgrad}(f + b^d),\ a,\ (W, \beta),\ e &\longrightarrow \tau, \delta, \phi \longrightarrow \text{rotated } q, k. \end{aligned}$$

In its radial-only specialization, the direct-pair backward has no angle data dependency or load. Conversely, the clock's m and a arguments are detached, so the phase VJP returns no gradient to the radial gate, radial bias, or $A^{\log}$. Unlike Mamba-3, RDiag does not add an angular-clock contribution to the dynamical radial-step gradient. Among the new phase-specific parameters, gradients reach the angle projection and clock weight/bias; they also return to the shared hidden states through both projections. The isolation claim is specifically that no clock edge reaches the radial parameters.

5 What to take from Mamba-3

The useful transfer from Mamba-3 is structural: encode rotation as a relative query/key phase with a small shared angle projection and a token/head clock. The estimator-specific pieces should remain KLA-specific.

Aspect	Mamba-2 SISO	Mamba-3 SISO	RDiag KLA E2
Transition geometry	Scalar radial decay; no rotation	Radial SSM plus relative pair rotation	Pair-tied radial KLA plus relative pair rotation
Clock	$DT_{th} > 0$	$DT_{th} > 0$ shared by decay and angle	$0 < \tau_{th} < 0.1$, based on detached live radial mean plus residual
Angle rows	None	Token/pair rows shared across heads	Token/pair rows shared across heads; head-specific τ
Phase gauge	None	Positive cumulative phase	Negative cumulative phase; sign documented by Eq. (26)
Write	Euler-like $DT\,B\,x$	Trapezoidal two-endpoint SSM write	Kalman innovation with anisotropic gain
Clock gradient	Only SSM dynamics	Angle dDT is added to dynamical dDT	Phase/radial VJPs intentionally isolated
Covariance	No Kalman posterior covariance	No KLA diagonal information update	Diagonal information in moving frame; pair-isotropic prediction parameters

Concretely, RDiag should take the following from Mamba-3:

1. represent rotation by the same adjacent-pair transform on queries and keys, avoiding a dense recurrent rotation;
2. use a bounded angular rate $\pi \tanh(e)$ and one token/head clock broadcast over pair modes;
3. share the compact angle projection across heads, while permitting head-specific timing;
4. accumulate phase and streaming carry in FP32, with a specified modular reduction contract;
5. treat every dependency of a shared clock explicitly in backward and test cancellation-heavy reductions against the executable reference; and
6. retain half- or full-prefix rotary layouts rather than forcing every key lane to rotate.

It should intentionally *not* copy the following:

1. Mamba-2's scalar-only transition as a purported rotational mechanism;
2. Mamba-3's trapezoidal SSM write, because KLA's innovation update defines a different estimator;
3. Mamba-3's angle-to-DT gradient coupling, because E2 deliberately uses a detached radial timescale;
4. approximate softplus, trigonometric, or reassociated reduction sequences without proving the required finite-precision parity;
5. independent radial coefficients inside a rotated pair while claiming a literal scalar-damped rotation; they instead define a more general rotated anisotropic contraction; or
6. hard clock clamps or straight-through boundary gradients, which conceal saturation instead of making it observable.

6 Implementation contract

6.1 Layout and forward pipeline

The production-oriented route is restricted to contiguous CUDA tensors, BF16 projected $q, k, v, f_{\text{raw}}, z_{\text{raw}}, y_{\text{raw}}$, FP32 frontend parameters, equal key/value head counts, and $K \in \{64, 128\}$.

The clock preactivation z^{clk} is a separate FP32 quantity. The rotary fraction is $1/2$ or 1, so K_r is positive and even. Adjacent memory lanes $(0,1),(2,3),\ldots,(K_r-2,K_r-1)$ form the rotary pairs; no split-half pairing is permitted.

The forward pipeline is:

1. project q,k,v and the diagonal-KLA radial/noise quantities;
2. compute the standard frontend's frozen α^{in} and ω^{in};
3. call the unchanged direct-pair operator with a detached angle, retain its pair-tied α, ω, and log-decay prefix, and discard its legacy phase output;
4. independently materialize the precise detached FP32 $[B,T,H,K]$ clock steps and form the exact native-framework radial mean m described below;
5. keep Wx as the existing FP32 GEMM, then fuse the pointwise clock chain and Eq. (24), using the exact CUDA-eager sigmoid quotient;
6. perform the bounded negative phase scan in chunks of 64, retaining an FP32 carry;
7. normalize q,k in FP32, round the memory-facing copies as specified, and rotate adjacent pairs by the same phase; and
8. run the diagonal information/gain kernel and the S3 chunked KLA memory recurrence.

The gain-facing key remains FP32. The memory-facing query/key route is rounded through BF16 in the same place as its executable reference before rotation/final rounding; changing that boundary is a numerical model change for parity purposes.

6.2 The precise detached-clock mean

The actual radial dynamics retain the FLA softplus used by the qualified baseline. The detached clock uses a separate precise FP32 copy

$$\mathrm{softplus}_{20}(x) = \begin{cases} x, & x > 20, \\ \log 1p(\exp x), & x \le 20, \end{cases} \tag{34}$$

evaluated with the precise device intrinsics. To reproduce the eager reference, the optimized path must materialize

$$\mathrm{clock\_dt} \in \mathbb{R}^{B\times T\times H\times K} \tag{35}$$

in FP32 and then invoke the framework reduction

$$\mathrm{radial\_clock} = \texttt{torch.mean(clock\_dt, dim=-1, out=radial\_clock)}. \tag{36}$$

A mathematically equivalent Triton lane sum is not an exact substitute. It changes addition order and was shown to leave small residual differences that can be amplified by long phase scans and cancellation-heavy angle gradients. The precise clock copy is detached; the FLA-softplus radial outputs and their VJP remain untouched. The temporary costs exactly $4BTHK$ bytes. At $B=4, T=4096, H=16$, that is 64 MiB for $K=64$ and 128 MiB for $K=128$; at $H=18, K=128$ it is 144 MiB. This cost belongs in the end-to-end memory and speed gate rather than being hidden by a numerically different reduction.

6.3 Reusing the direct-pair radial path safely

The production wrapper does not maintain a second implementation of Eq. (19). It calls the established direct-pair transform with `angle_raw.detach()`, ignores the returned legacy delta, and reuses only g, α, and ω. This keeps the radial forward arithmetic and its VJP identical to the already-qualified direct-pair path.

Discarding the legacy delta makes its incoming cotangent absent. The direct-pair backward turns that fact into the compile-time specialization `HAS_DDELTA=false`. In this specialization the generated kernel does not load the detached angle, evaluate its tanh, or form the phase term $\pi \tanh(e)\,\overline{\delta}$. The guard is semantically important: merely setting $\overline{\delta} = 0$ would not isolate a nonfinite discarded angle, because IEEE arithmetic gives $0 \times \mathrm{NaN} = \mathrm{NaN}$. The phase-enabled `HAS_DDELTA=true` algebra is unchanged, while dedicated NaN/$\pm\infty$ tests establish that a discarded phase cannot contaminate radial outputs or gradients.

6.4 Exact sigmoid and bounded-delta forward

Let

$$\ell_{bth} = \log m_{bth} - \log(C - a_h) + \tanh(z^{\mathrm{clk}}_{bth}). \tag{37}$$

For FP32 CUDA inputs, PyTorch's native sigmoid kernel evaluates $1/(1 + \exp(-\ell))$. The generic Triton `tl.sigmoid` uses a different approximate lowering and produced one-ULP clock differences even when ℓ was bitwise identical. The production kernel therefore spells out the eager expression as

$$\tau^{\mathrm{raw}}_{bth} = 0.1\,\texttt{tl.div\_rn}\left(1,\ 1 + \texttt{libdevice.exp}(-\ell_{bth})\right), \tag{38}$$

then applies the fail-visible validity predicate to obtain τ. The angle nonlinearity is computed once by native ATen as `torch.tanh(angle_raw.float()).contiguous()` and passed to the fused clock kernel, which forms $\delta = \pi \tanh(e)\tau$. This boundary makes the actual-logit τ and δ bitwise equal to the eager oracle, rather than merely close in real arithmetic.

6.5 Phase scan and backward order

The phase forward is an inclusive FP32 scan. Each chunk computes $\phi = \mathrm{carry} - \mathrm{cumsum}(\delta)$, wraps valid values to $[0, 2\pi)$, and passes the wrapped final value as the next carry. Before reduction, $|\phi|$ must not exceed 65,536; nonfinite, out-of-contract, or noncanonical residues become sticky NaNs. Its VJP is the global negative suffix scan

$$\overline{\delta}_t = -\sum_{u=t}^{T} \overline{\phi}_u, \tag{39}$$

with the final-carry cotangent subtracted from every token when present.

For the fused clock/delta backward, operation order is part of the contract. Let $g_{thj} = \overline{\delta}_{thj}$. The eager-compatible order is

$$\overline{\tau}_{th} = \sum_j \left[g_{thj} \big(\pi \tanh(e_{tj}) \big) \right], \tag{40}$$

$$\overline{z}^{\mathrm{clk}}_{th} = \mathrm{mask}_{th}(\overline{\tau}_{th})\, \tau^{\mathrm{raw}}_{th} \left(1 - \frac{\tau^{\mathrm{raw}}_{th}}{C} \right) \left(1 - \tanh^2 z^{\mathrm{clk}}_{th} \right), \tag{41}$$

$$\overline{e}_{tj} = \left[\left(\sum_h g_{thj} \tau_{th} \right) \pi \right] \left(1 - \tanh^2 e_{tj} \right). \tag{42}$$

Here $\mathrm{mask}_{th}(g)$ is g when the forward validity predicate selected τ_{th}^{raw} and zero when it selected the detached NaN branch. The subsequent multiplication order is retained, so a NaN local derivative remains fail-visible even after a zero cotangent mask. Equation (40) must be evaluated as the displayed broadcast multiply followed by the framework's sum over P; pulling π outside the sum is not accepted as "equivalent." For Eq. (42), multiply by τ, reduce over H, multiply by π, and only then apply the tanh derivative. Concretely, production saves the native-ATen tanh output from the forward and evaluates

```
torch.ops.aten.tanh_backward( (d_delta * tau[..., None]).sum(dim=2) * pi, angle).
```

The result is cast to `angle_raw`'s dtype (BF16 on the production route) before the angle projection VJP. No Triton angle-partial tensor or head-reduction kernel remains. The validity mask is applied to the incoming clock cotangent before the sigmoid/tanh clock chain, matching the eager `where(valid, raw_tau, detached_nan)` VJP. Weight and hidden-state gradients then flow through the ordinary linear operations.

6.6 Persistent state and FSDP

The anchor a_h must satisfy all of the following:

1. FP32, contiguous, shape $[H]$, finite, and strictly inside $(0, 0.1)$;
2. registered as a persistent buffer and present in the model state dictionary;
3. absent from named parameters and optimizer groups;
4. identical across distributed ranks after construction, FSDP wrapping, save, and resume;
5. restored from the checkpoint rather than regenerated from the current radial bias.

Production enforces this with `bounded_rdiag_fsdp_mixed_precision`: it retains the installed BF16-mixed parameter/reduction policy but overrides `buffer_dtype=torch.float32`. Resume tests compare the anchor bytes before and after round trips and verify that fresh construction cannot silently overwrite a checkpointed anchor.

6.7 Initialization and health signals

The angle projection weight and bias, and the bounded-clock residual weight and bias, are initialized to zero. Rotation therefore starts at the *pair-tied* diagonal-KLA function: $e = 0$ gives $\delta = 0$ even when τ is nonzero, while the pair averaging in Eq. (19) remains active. This is preferable to initializing the clock at zero, which would place it on a boundary and kill useful clock gradients once angles become live.

The audited bounded RDiag named configurations use $r_{\min} = 0.01$, process-noise floor $\omega_{\min} = q_{\min} = 0$, and `gain_init=iso`. The last setting zeroes the output map of the process-noise projection, so ω starts uniform across key channels; it does *not* initialize ω itself to zero, because the positive softplus remains. These are recipe and initialization choices rather than changes to Eqs. (28)–(33).

At every checked optimizer boundary, diagnostics should aggregate across microbatches, layers, and ranks:

- nonfinite counts for the radial mean, base logit, residual, logit, raw clock, valid clock, derivative, phase, parameters, and gradients;
- exact endpoint and near-boundary counts, using 10^{-4} and $0.1 - 10^{-4}$ as the observed lower and upper near-boundary bands;

- minimum positive clock Jacobian and a near-dead count below 10^{-8};
- token variation of the residual projection after bootstrap;
- phase magnitude, delta magnitude, and canonical wrapped-range checks; and
- schema/count agreement across ranks, including exactly one observation per expected micro-batch/layer pair.

The kernel itself fails visibly at invalid endpoints. Promotion screens may impose stricter distributional thresholds than the continuous health controller; both classes of gate are needed.

7 Reference pseudocode

The following pseudocode fixes the mathematical order without prescribing launch geometry.

1. **Radial and clock preparation.**

$$\begin{aligned}
x^d &= f + \texttt{dt_bias}, \\
d^{\text{radial}} &= \texttt{FLA_softplus}(x^d), \\
\texttt{clock_dt} &= \operatorname{softplus}_{20}(\operatorname{stopgrad}(x^d)) \quad \text{materialized FP32 } [B, T, H, K], \\
m &= \texttt{torch.mean(clock_dt, dim=-1, out=radial_clock)}.
\end{aligned}$$

   Invoke the unchanged direct-pair transform with a detached angle, pair-average d^{radial} and ω on the rotary prefix, and construct α from Eq. (19). Discard its legacy delta and do not use `clock_dt` for radial dynamics.

2. **Bounded clock and phase.**

$$\begin{aligned}
z^{\text{clk}} &= \texttt{linear}(x, W)/\sqrt{D} + \beta, \\
\ell &= \log m - \log(0.1 - a) + \tanh z^{\text{clk}}, \\
\tau &= 0.1\,\texttt{div_rn}(1, 1 + \texttt{libdevice.exp}(-\ell)), \\
\widehat{e} &= \texttt{torch.tanh}(e\texttt{.float()}), \\
\delta &= \pi \widehat{e}\,\tau, \\
\phi_t &= \operatorname{wrap}_{2\pi}(\phi_{t-1} - \delta_t).
\end{aligned}$$

   Reject invalid open-clock or phase-contract values by producing observable NaNs.

3. **Addresses and Kalman recurrence.** Rotate the same adjacent prefix of normalized q and k by $R(\phi)$, preserve tail lanes, then apply Eqs. (28)–(33). The chunked implementation must match this token recurrence under the documented BF16/FP32 boundary.

4. **Backward.** Run Eq. (39); use Eqs. (40)–(42) exactly; evaluate the angle derivative with native `aten.tanh_backward` on the saved $\widehat{e}$; compile the discarded direct-pair phase branch with `HAS_DDELTA=false`; return no clock gradient to m or a; and run the independent radial/Kalman VJPs.

8 Lessons from the E-screen and performance/numerical debugging

An open interval is not a distributional guarantee. An earlier bounded-clock E screen completed 519 optimizer updates and produced no exact endpoints or nonfinite clocks, yet the step-520 evaluator rejected it. In layer 0, the lower-near-boundary fraction was 0.00205326, above the preregistered 10^{-3} threshold; the observed clock range was approximately $[8.65 \times 10^{-5}, 0.09813]$. Thus a smooth sigmoid bound alone did not prevent probability mass from accumulating near a boundary. The design response was the detached live radial base plus a bounded token/head residual, not a harder clamp.

The redesigned E2 screen reached its step-520 evaluation after 519 completed optimizer updates (520 measured gradients), with zero lower and upper near-boundary counts in the reported gate and an observed clock range of approximately [0.01060, 0.08181]. That result supports the clock design and justified integration work. It does not prove long-run training quality, kernel parity under every shape, distributed resume correctness, or a production speedup.

Correctness-first composition exposed the actual performance target. At $B = 4, T = 4096, H = 16, K = V = 128, D = 2048$, the unfused rotational-v1 wrapper measured 24.706 ms for combined mixer forward/backward versus 13.735 ms for its frozen control, a 1.799× ratio (79.9% overhead). The identical-input frozen S3 stage was 0.999× its control, localizing the cost to the eager second-dt, pair tying, phase/trigonometry, three rotations, and prefix rebuild. This motivated the direct radial kernel, fused normalization/rotation, and fused clock/delta path while retaining the established gain and S3 kernels. These are historical isolation numbers for the v1 geometry, not a speed claim for the final E2 composition. The later final A/B measurements, including the exact-mean FP32 scratch, are reported in Table 1.

A tiny local error can become a large gradient discrepancy. The first optimized clock path reused the FLA radial softplus. Against eager PyTorch, the initial radial-mean discrepancy was only about 6.33×10^{-8} in maximum absolute value and 1.54×10^{-6} in relative L_2. After the logit, long phase scan, BF16 rotation boundary, and cancellation across heads, the angle-weight gradient showed about 3.81×10^{-3} relative L_2 error in the focused diagnostic. Near-zero reference entries also made elementwise relative metrics misleading, so absolute, normwise, and cancellation-stress gates are all required.

Using Eq. (34) reduced the clock error substantially, but a Triton reduction still did not reproduce eager bitwise. Materializing the FP32 $[B, T, H, K]$ precise values and using the exact framework mean was bitwise identical in the focused $K = 64$ and $K = 128$ experiment while leaving the FLA radial α, log-prefix, and ω outputs bitwise unchanged. Likewise, algebraically reassociated all-Triton backward reductions failed the focused VJP gate; the explicit framework reduction of Eq. (40) and the factor order of Eq. (42) passed it. The lesson is narrow but important: the executable eager graph, including reduction order, is the reference for this mixed-precision path.

Elementary functions are also part of the executable reference. After the exact mean fix, a full $K = 64$ trace found bitwise-identical radial clock, base logit, bounded residual, and final logit, yet the fused clock still differed from eager at 61,655 of 262,144 FP32 τ entries (maximum absolute error 7.45×10^{-9}). Those one-ULP clock differences propagated to 929,205 of 4,194,304 delta entries and were amplified by the length-4096 phase scan. The cause was `tl.sigmoid`, not the model equations. Replacing it with Eq. (38), matching PyTorch's CUDA source expression and rounded division, removed the actual-logit discrepancy. Computing the angle tanh and its backward with native ATen then removed the remaining reduction/transcendental-order drift. This is why the final path uses selective eager operations at two small boundaries rather than insisting that every operation be fused into Triton.

9 Invariants and qualification status

The following are invariants, not tunable approximations:

1. rotary prefix width is positive and even; pairing is adjacent;
2. α and ω are exactly equal within each selected pair;
3. q and k receive the same phase and same rotation matrix;
4. the phase scan is negative and inclusive, and its backward is a negative suffix scan;
5. m and a are detached FP32 quantities; phase gradients do not enter radial dynamics;
6. the valid clock and its local derivative are finite and strictly positive, with $0 < \tau < 0.1$;
7. the anchor is persistent FP32 checkpoint state, not optimizer state;
8. precise clock softplus plus `torch.mean` is separate from FLA radial softplus;
9. the bounded sigmoid uses Eq. (38), not `tl.sigmoid`;
10. the displayed forward and backward reduction orders are preserved, with the angle tanh and its VJP evaluated by native ATen;
11. a discarded direct-pair delta selects `HAS_DDELTA=false`, eliminating every angle access and phase product from the radial backward; and
12. the unrotated tail is an exact pass-through of diagonal KLA.

The one-H200 kernel/layer checklist is closed by the cumulative qualification campaign:

1. pure-PyTorch token recurrence versus optimized forward for zero, half, and full rotation;
2. first-order VJPs for every input/parameter, including adversarial cancellation and partial chunks, at $K = 64$ and $K = 128$;
3. exact checks for radial outputs and detached clock mean where exactness is part of the contract, plus justified tolerances elsewhere;
4. segmented/streaming equivalence, phase carry, wrap-boundary, and sticky-NaN tests;
5. zero-angle equivalence to pair-tied diagonal KLA and angle-sign/gauge unit tests;
6. multi-step health checks covering bootstrap-zero angle, subsequently live clock gradients, token variation, and no nonfinite/endpoints;
7. exact full-layer forward/VJP comparison with the sealed E2 candidate at $K = 64$, $K = 128$, and the 18-head 1.3B-style geometry; and
8. interleaved A/B timing against both the sealed E2 candidate and the legacy pair clock.

The authoritative one-H200 job 1777777 completed with exit code 0 and reported 117 passing core RDiag tests followed by 54 passing E2 tests. Table 1 gives the paired runtime ratios for an exact RDiag-layer forward plus backward; values below one are faster than the named reference. Every upper endpoint of the bootstrap 95% confidence interval is below the 1.05 promotion limit.

Thus the optimized E2 implementation is bitwise faithful where exact parity is required, faster than the sealed accepted implementation in all three measured geometries, and within roughly 1–1.4% of the simpler legacy clock. This closes the E2 one-H200 kernel/layer gate, but not a full production-submission gate. Train-smoke, distributed checkpoint/resume, and multi-node

Table 1: Final H200 paired A/B timing from job 1777777, with 40 measured rounds after 20 warmups. "Accepted" is the sealed eager E2 candidate and "legacy" is the production pair-dt path.

Geometry (H, D, K)	E2/accepted	95% CI high	E2/legacy	95% CI high
$(16, 1024, 64)$	0.969512	0.978247	1.014135	1.020172
$(16, 2048, 128)$	0.950407	0.956546	1.008590	1.013666
$(18, 2304, 128)$	0.960111	0.964929	1.013622	1.014984

soak/attestation remained for that baseline at the time of sealing. The next section records the later shared-DT candidate and its separate, still-open qualification.

10 Shared-DT engineering status as of September 3, 2026

This section records a newer production candidate and does not revise the closed E2 result above. The candidate starts from commit `6472a0711eb61177c3ef91845855fffa82202290` and replaces E2's detached bounded clock with a shared-DT transition. Its qualification is *not complete*: four gates passed in the first campaign, the gradient gate exposed both a defective acceptance rule and real rotational-gradient amplification, the soak was cancelled before allocation, and no aggregate authorization was built.

10.1 Relation to Mamba-2 and Mamba-3

The source-level comparison in Sections 2 and 3 found no discrepant transition equation or kernel sign bug in the shared-DT path. Its radial factor is the Mamba-2 form

$$\alpha_{th} = \exp[-\exp(A_{\log,h}) DT_{th}]\,, \tag{43}$$

and its bias-free pair-angle projection follows the installed Mamba-3 construction,

$$\delta_{thj} = \pi \tanh(e_{tj}) DT_{th}, \qquad \phi_{thj} = \sum_{s \le t} \delta_{shj}, \tag{44}$$

with one angle value per token/pair broadcast over heads, a headwise DT, a cumulative phase scan, and the corresponding reverse-suffix backward. The apparent phase-sign difference is only the gauge $W_{\text{angle}} \mapsto -W_{\text{angle}}$ and therefore cannot explain a gradient-norm change. The relevant difference from the E2 model is instead that the angle and DT paths remain live and the same shared DT controls radial decay and rotation.

10.2 Issue, diagnosis, and completed work

Two distinct problems were found. First, the original per-microbatch global-gradient limit of 10 was not a valid gate: all 24 measured model/batch observations exceeded 10, including stock DiagKLA at 14.001–14.232. Stock DiagKLA and the bounded-clock layer were also architecture-mismatched controls. Second, the shared angle projection creates genuine coherent amplification when its output is broadcast over all recurrent heads and differentiated through the cumulative rotation. This is a model-parameterization issue, not evidence of a faulty radial or phase kernel.

On the same eight real FineWeb batches, the unmodified shared-DT model produced global norms 34.596–45.300 (median 38.758), versus a DiagKLA median of 14.096. Disabling rotation returned the median to 14.180, while detaching only the angle path gave 18.035 and zero angle initialization still gave 25.593. A phase-only DT cap of 0.1 was insufficient. Scaling the effective angle preactivation by $1/\sqrt{16} = 0.25$ gave median 16.765 and shared/Diag median and 95th-percentile nearest-rank ratios 1.191 and 1.261, with live angle and DT gradients. These interventions isolate head-shared cumulative rotation as the amplification source.

Completed engineering work. The Mamba-2/Mamba-3 source audit, exact H200 measurement, rotation/gradient ablations, matched- backbone control, and scale/cap sweep are complete. The replacement design and implementation plan received an independent review with no remaining Important findings. In campaign `9870d362bcbb4ec3b9b3cafb13459ef8`, production-H200 (job 1824028), train-smoke (1824365), checkpoint-fault (1824367), and authenticated-resume (1824368) produced recursively valid seals with SHA-256 prefixes `ecc03259`, `9ca6354e`, `06c1f60a`, and `656c9412`, respectively. Gradient job 1824370 failed the invalid absolute-10 rule and produced no seal; the soak was cancelled before allocation. These seals characterize the unmodified base commit only and do not qualify the proposed head-scaled change.

Diagnostic evidence. The primary metric artifacts are

- `diagnostic_runs/gradient_norm_6472a071_20260902T170943Z/metrics.json`, SHA-256 `9c6a8a96a297799a`;
- `diagnostic_runs/gradient_rotation_ablation_6472a071_20260902T183817Z/metrics.json`, SHA-256 `274abd1902e7c7d3`;
- `diagnostic_runs/gradient_matched_control_6472a071_20260902T185000Z/metrics.json`, SHA-256 `d908637092fdfc3e`; and
- `diagnostic_runs/gradient_angle_scale_6472a071_20260902T185100Z/metrics.json`, SHA-256 `e503ab386a9de79d`.

All are rooted at `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1`.

10.3 Approved implementation and remaining qualification

Chosen change. For H recurrent heads, the layer will derive the nonpersistent scalar $H^{-1/2}$ and form

$$e^{\text{eff}}_{tj} = H^{-1/2}(W_{\text{angle}} x_t)_j, \qquad \delta_{thj} = \pi \tanh(e^{\text{eff}}_{tj}) DT_{th}. \tag{45}$$

The projection remains bias-free and its stored weight is unchanged. In the BF16 production path, the layer performs the multiply on the BF16 projection output before the frontend's FP32 cast and FP32 `tanh`; the frontend must not scale a second time. The scale is neither a parameter nor a buffer, so old state dictionaries remain strictly loadable. This preserves the transition family and both learning paths while normalizing the initial head-summed VJP.

The replacement gradient gate is an exact same-model/same-state two-arm intervention. The `shared_dt` arm runs normally; `shared_dt_no_rotation` replaces only each angle- projection output by the connected zero `output * 0.0`. Config, full state, parameter inventory, and radial shared-DT path must be identical. Acceptance requires shared/no-rotation nearest-rank median ≤ 1.25 and 95th percentile ≤ 1.50, both inclusive. The microbatch runaway sentinel becomes an inclusive 100; the late-training soak limit of 10 is unchanged. Only the gradient sample/result/recipe/seal closure advances to schema v3. Aggregate-v3 retains its schema name but must recursively require the v3 gradient seal.

Active checkout and plans. Implementation is isolated from the main checkout in this worktree:

> `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_worktrees/rdiag_shared_dt_headscale_20260902`.

The observed checkout is branch `codex/rdiag-shared-dt-angle-headscale-20260902`, based on the commit above. The reviewed documents are

- `docs/plans/2026-09-02-rdiag-shared-dt-angle-headscale-design.md`; and
- `docs/plans/2026-09-02-rdiag-shared-dt-angle-headscale-plan.md`.

The primary implementation and qualification paths are

- `lit_gpt/rdiag_shared_dt_kla.py`;
- `lit_gpt/kla_ops/rdiag_shared_dt.py` and `lit_gpt/kla_ops/rdiag_bounded_phase.py`;
- `scripts/tests/rdiag/test_rdiag_shared_dt_production_host.py` and `scripts/tests/rdiag/test_rdiag_shared_dt_production_h200_v1.py`;
- `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab.py` and `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab_run.py`; and
- `scripts/train/rdiag_kla/rdiag_recipe.py`, `scripts/tests/rdiag/attest_rdiag_shared_dt_gate.py`, and `scripts/train/rdiag_kla/rdiag_shared_dt_qualification.py`.

Remaining work. The head scale and repaired gradient closure are designed but not yet implemented or qualified. The remaining sequence is:

1. add failing host contracts, then implement the exactly-once layer-side scale and unchanged state-schema behavior;
2. prove production-layer forward and all VJPs against the literal oracle at H=16, T=4096, BF16 and at H=18, while preserving radial outputs and live angle/DT/hidden gradients;
3. implement the same-state rotation-off runner, v3 gradient schemas, policy boundaries, readers, catalog digests, stale-schema rejection, and recursive aggregate-v3 validation;
4. pass focused and broad host/static gates, freeze a clean exact commit, and obtain separate exact-SHA specification/math and code-quality reviews;
5. push that reviewed commit, materialize a fresh immutable snapshot, and rerun all six gates: production-H200, train-smoke, checkpoint-fault, authenticated-resume, the real eight-batch gradient gate, and production soak; and
6. build aggregate-v3 only after all six seals validate, then submit the 750M run with the full standard-plus-all-configured-NIAH post-training evaluation tail.

Until that sequence closes, neither a production authorization nor a replacement training result is claimed.

A Reproducibility audit

RDiag implementation baseline. The sealed E2 baseline implementation statements in this note were audited against branch `rdiag` at Git commit `c3346feaecae1486cdbc71f82b4edeacc4198950` (tree `88e423d2926d8ae81945c08393a59861d285907c`). This pin identifies the sealed implementation snapshot; the present documentation-only working-tree update is intentionally outside that snapshot. Relevant regions are:

- `lit_gpt/rdiag_kla.py`, construction and buffers around lines 137–193, phase construction around 222–293, and the fused layer route around 409–516;
- `lit_gpt/kla_ops/rdiag_clock.py`, anchor and eager oracle around 72–119 and 190–270, exact fused clock around 283–390, and native-order custom backward beginning around line 515;
- `lit_gpt/kla_ops/rdiag_direct.py`, the unchanged direct-pair forward around line 162, guarded direct-pair backward around line 338, precise clock materializer around line 300, and radial wrapper around line 920;
- `lit_gpt/kla_ops/rdiag_phase.py`, negative scan contract and reference around 1–19 and 294–328;
- `lit_gpt/kla_ops/rdiag_bounded_phase.py`, bounded fail-visible phase contract around 1–11 and 41–172;
- `lit_gpt/kla_ops/rdiag_norm.py`, public fused normalization/rotation contract around 10–90;
- `lit_gpt/kla_ops/rdiag_norm_reference.py`, normalization, BF16 boundary, and adjacent-pair rotation around 24–96;
- `lit_gpt/kla_ops/diag_naive.py`, KLA equations around 14–32, 144–199, and 451–491;
- `lit_gpt/diag_kla.py`, process-noise and isotropic-output initialization around 280–300;
- `lit_gpt/config.py`, bounded RDiag recipe fields around 349–369 and 742–763;
- `lit_gpt/rdiag_training_health.py`, health thresholds and observed counters around 24–37 and 121–255, and fail-closed decision logic around 579–612;
- `lit_gpt/fsdp_strategy.py`, the FP32-buffer mixed-precision override beginning around line 131;
- `scripts/tests/rdiag/test_rdiag_bounded_h200.py`, exact radial, actual-logit, exceptional-value, full-layer parity, and A/B performance gates;
- `sanboxes/kernels/rotational_diag_kla_v1/DESIGN.md`, moving-frame semantics and the pair-tied literal model; and
- `sanboxes/kernels/rotational_diag_kla_v1/RESULTS.md`, the correctness and v1 performance evidence around lines 128–166.

The whole tree is sealed by the snapshot manifest identified below, which is the authoritative byte-level inventory. Three central files in the final parity implementation and gate have SHA-256 values

- `lit_gpt/kla_ops/rdiag_clock.py`,
  `d02a3f0d0994067afc174f1a77f7ebd8fcecd73c3a18956e72ab79ab49afc966`;
- `lit_gpt/kla_ops/rdiag_direct.py`,
  `9fa3e29102bfa512ecbc57fc04ce6453e3a33e7aed61e7049026b71fe81faace`;
- `scripts/tests/rdiag/test_rdiag_bounded_h200.py`,
  `68d7db9b6793a33ef2574c7a39527306c3b3b500a03e1807790ce59889be8517`.

The final result followed three deliberately narrow diagnoses. Commit `fe1b1258` made the radial clock mean bitwise exact but left three full-layer angle-weight mismatches in job 1777082. Job 1777229 then proved that the first remaining forward divergence occurred at sigmoid output despite a bitwise-identical input logit. Commit `4ce6e19` matched the CUDA-eager sigmoid and angle VJP, and commit `c3346fe` added the missing-delta compile-time guard that makes the detached radial path immune to nonfinite discarded angles.

Installed Mamba implementation. The source audit used `mamba-ssm 2.3.2.post1` installed under the active `hdn` environment. Paths and SHA-256 identities are:

- `mamba_ssm/modules/mamba2.py`, `605e4439ff0baec8d8acaf4a191d9f0570eea9900065a065909124c472b08707`; inspect roughly lines 95–136, 210–260, and 309–318;
- `mamba_ssm/modules/mamba3.py`, `de1b104865c941eb6eeac63c3a75d5913fc2e7a119d52991590ef19c381b05ac`; inspect roughly lines 76–87, 150–178, and 220–237;
- `mamba_ssm/ops/triton/mamba3/angle_dt.py`, `7d99f8cff37371cfbfe9d4dd36f17a16135fb07f2c6e14add3ad4dd5c0034e2e`; inspect roughly lines 83–118 and 298–337;
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_combined.py`, `410ad57555b5841fba7bd76fdafdc3ddf82741176489c3e3493f5c6f3c0c6012`; inspect roughly lines 90–105 and 247–259; and
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_fwd.py`, `68d8d0dba0e3913f2e996862603289fd0d7157e7dc57c872e04234be8bb82695`; inspect roughly lines 256–332, 348–352, and 386–425.

The hashes, package version, and paths are the durable identity; line numbers are navigation hints.

Qualification evidence and present status. The lower-tail rejection is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/probe_v1_1774423/probe_trace.json`. The redesigned E2 decision is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/qualification_bounded_de_20260828_1145/D_E_DECISION.json`. The precision experiment is recorded in `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/radial_clock_precision_1776573/result.json`; focused VJP evidence is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_vjp_diag_k64_1776509.log`, and the five-case composed check is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_angle_order_1776788.log`. These artifacts support the specific debugging observations in Section 8.

The sealed implementation snapshot used by the one-H200 gate is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/snapshots/production_e2_20260828_2344`. Its `SOURCE_MANIFEST.json` SHA-256 is

`05db76db18b8f4fe170a102e2e5df28bb54a4e36ca7ebe4565a1d813b44530a3`.

The authoritative qualification log is `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_h200_1777777.log`. SLURM job 1777777 completed with status `COMPLETED`, exit code `0:0`, 117 passing core tests, and 54 passing E2 tests. This closes the one-H200 kernel/layer gate. Distributed training-system checks and the multi-node soak/attestation remain before submission; training completion and the full post-training evaluation are subsequent empirical deliverables.